SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)*

Assessment Tool Weightage: 50%

Dr.T.P.Anithaashri

QUESTIONS: 10 Total Marks: 50

Annexure A

RL Basic Experiments demo with Keras and TensorFlow using Anaconda navigator

Duration: 2hrs

Code of Conduct:

I K.Vijay Bhaskar Reddy(192425155) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Experiment 1

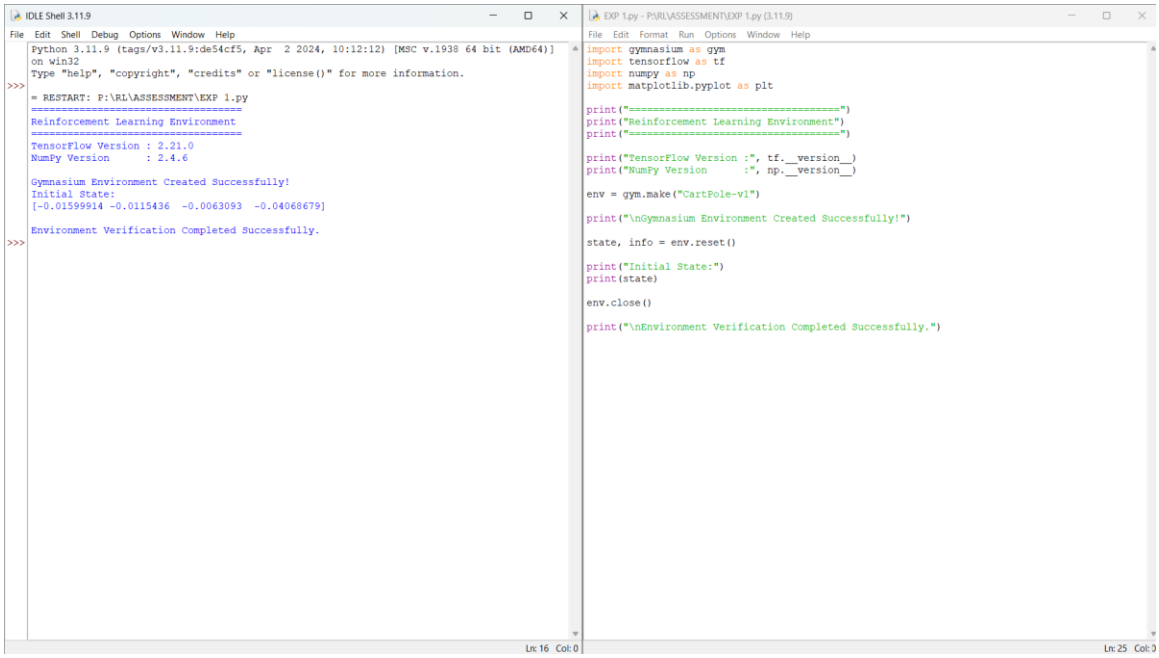
Title: Installation of Reinforcement Learning Development Environment

Aim: To install and configure Python, TensorFlow, Keras, Gymnasium, NumPy and Matplotlib and verify the RL environment.

Algorithm:

- Import the required libraries.
- Initialize the environment/model.
- Execute the required RL procedure.
- Display the output.
- Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 2

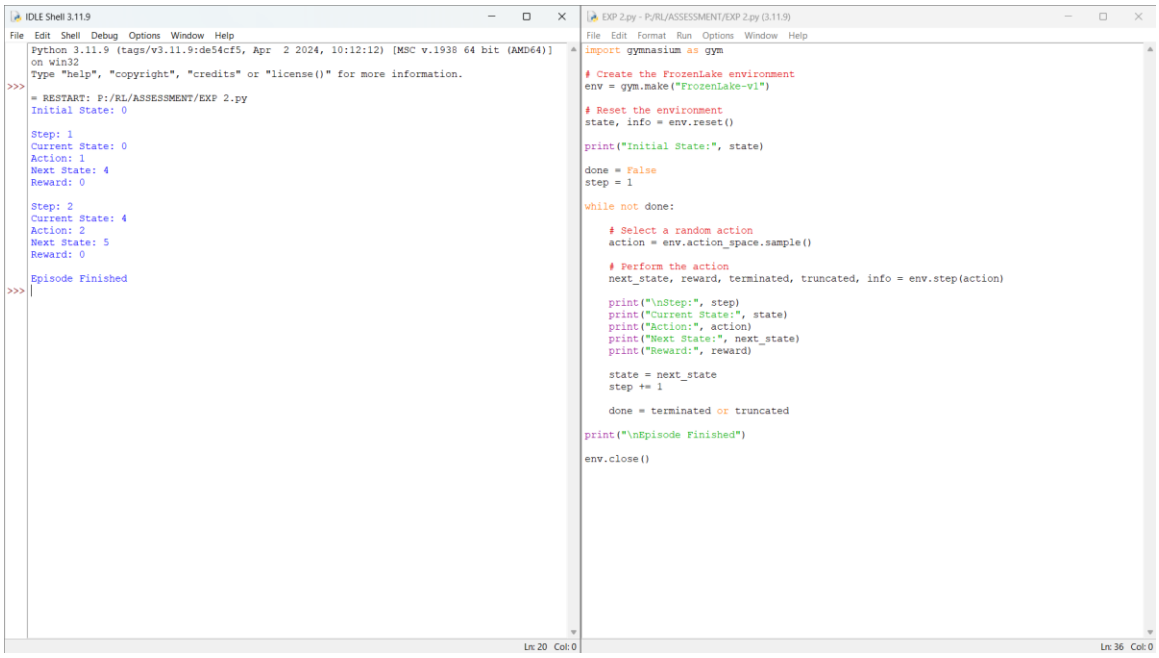
Title: Familiarization with OpenAI Gym/Gymnasium Environments

Aim: To create and interact with the FrozenLake environment and understand states, actions, rewards and episodes.

Algorithm:

- Import the required libraries.
- Initialize the environment/model.
- Execute the required RL procedure.
- Display the output.
- Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 3

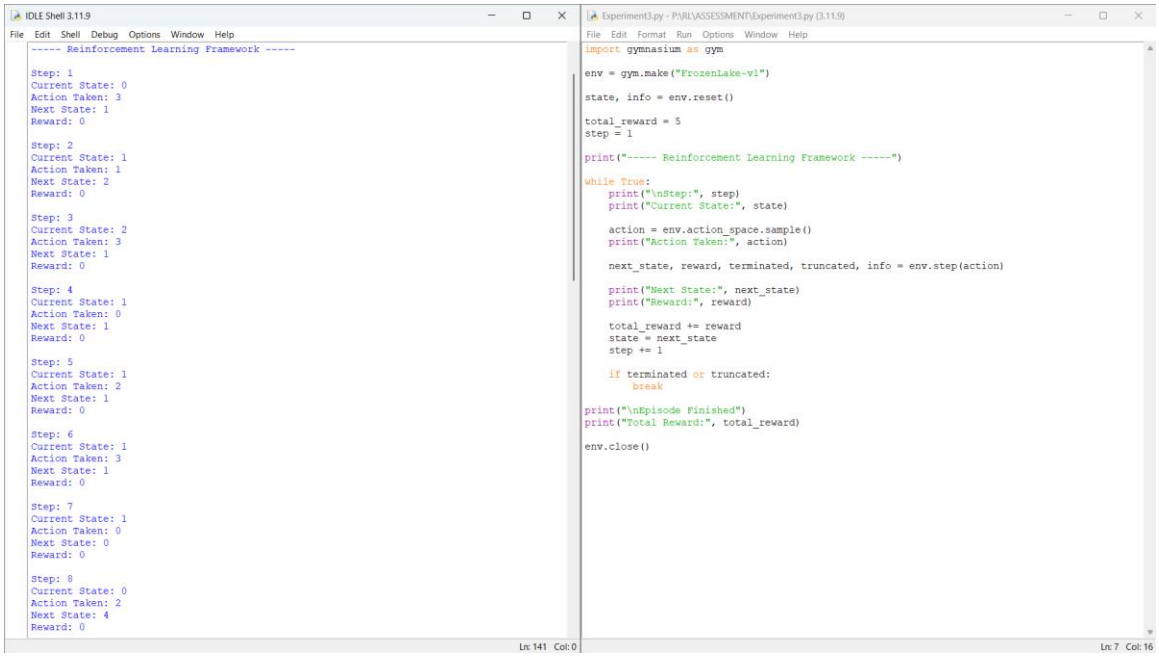
Title: Implementation of Reinforcement Learning Framework

Aim: To implement the basic RL framework using an agent interacting with the environment.

Algorithm:

- Import the required libraries.
- Initialize the environment/model.
- Execute the required RL procedure.
- Display the output.
- Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 4

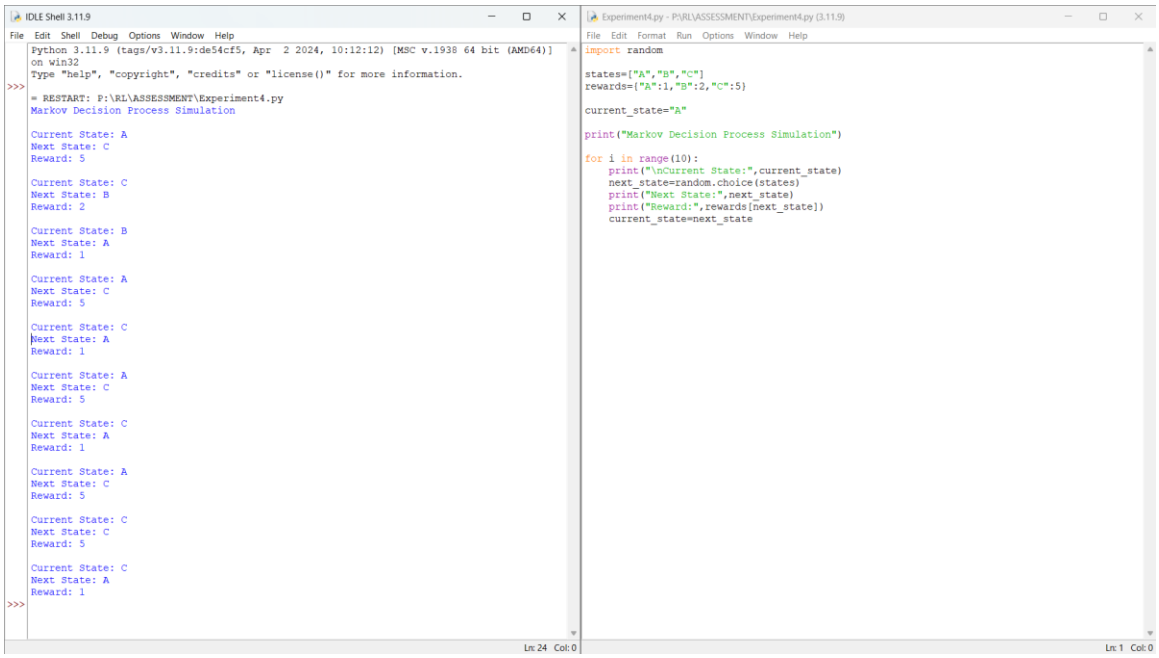
Title: Simulation of Markov Decision Process (MDP)

Aim: To simulate state transitions, rewards and policies using a simple MDP.

Algorithm:

- Import the required libraries.
- Initialize the environment/model.
- Execute the required RL procedure.
- Display the output.
- Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 5

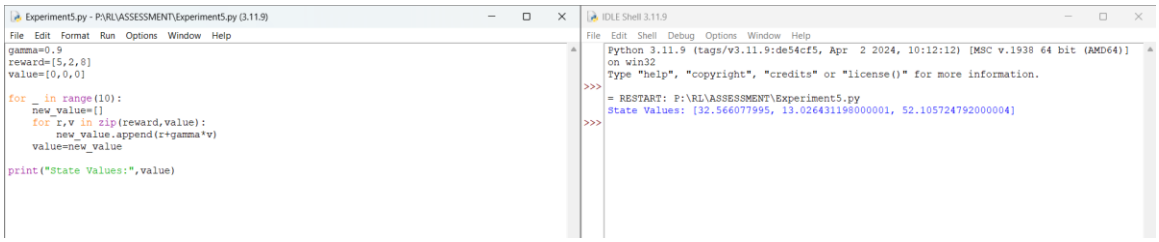
Title: Bellman Equation Demonstration

Aim: To implement Bellman Expectation and Bellman Optimality equations.

Algorithm:

- Import the required libraries.
- Initialize the environment/model.
- Execute the required RL procedure.
- Display the output.
- Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 6

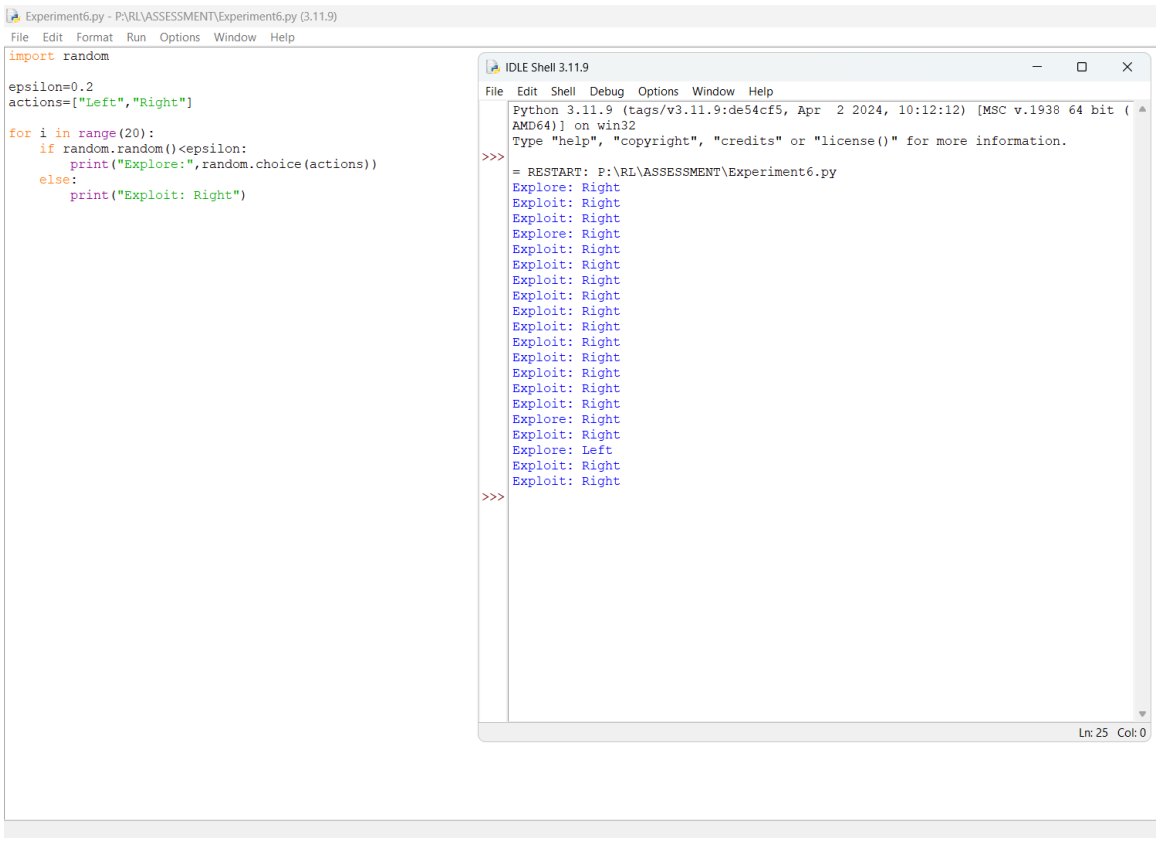
Title: Exploration vs Exploitation

Aim: To compare Greedy, ε-Greedy and Random strategies.

Algorithm:

• Import the required libraries.
• Initialize the environment/model.
• Execute the required RL procedure.
• Display the output.
• Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 7

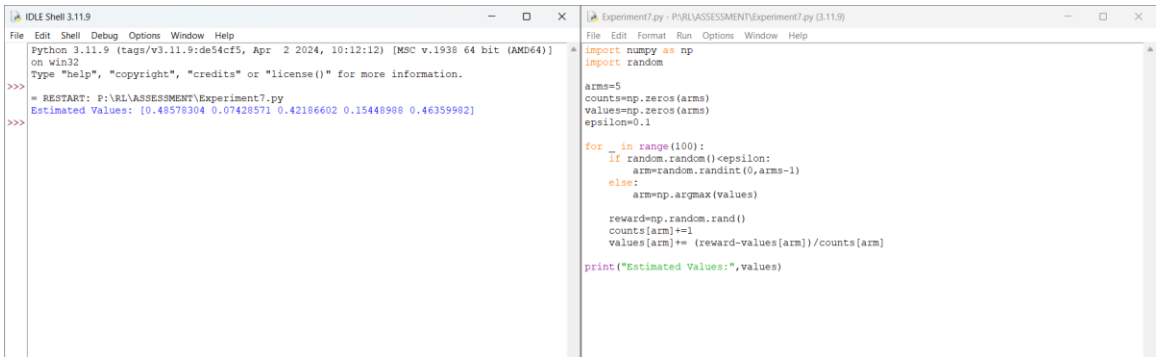
Title: Multi-Armed Bandit Problem

Aim: To implement the Multi-Armed Bandit problem using ε-Greedy.

Algorithm:

• Import the required libraries.
• Initialize the environment/model.
• Execute the required RL procedure.
• Display the output.
• Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 8

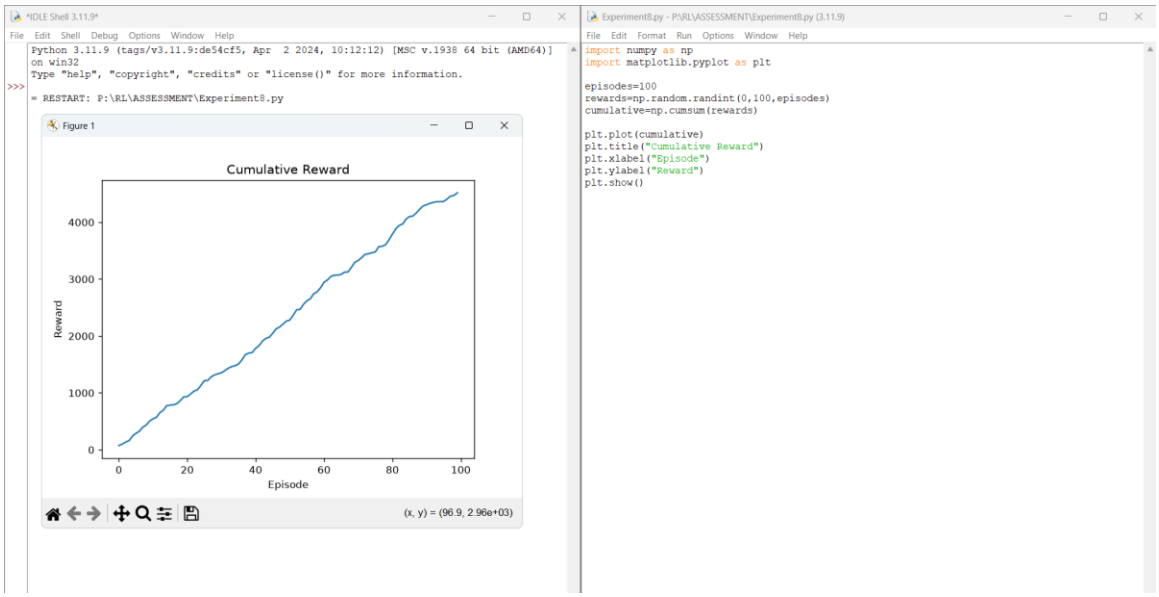
Title: Reward Visualization

Aim: To visualize cumulative rewards using Matplotlib.

Algorithm:

• Import the required libraries.
• Initialize the environment/model.
• Execute the required RL procedure.
• Display the output.
• Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 9

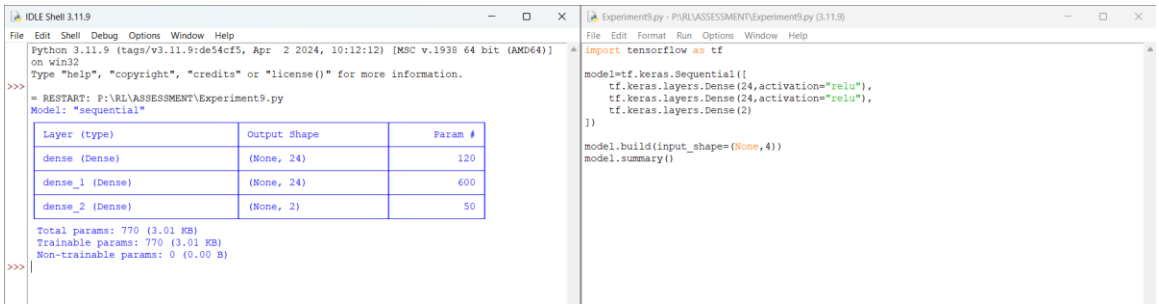
Title: TensorFlow and Keras Neural Network

Aim: To build a simple feed-forward neural network for RL.

Algorithm:

• Import the required libraries.
• Initialize the environment/model.
• Execute the required RL procedure.
• Display the output.
• Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.

Experiment 10

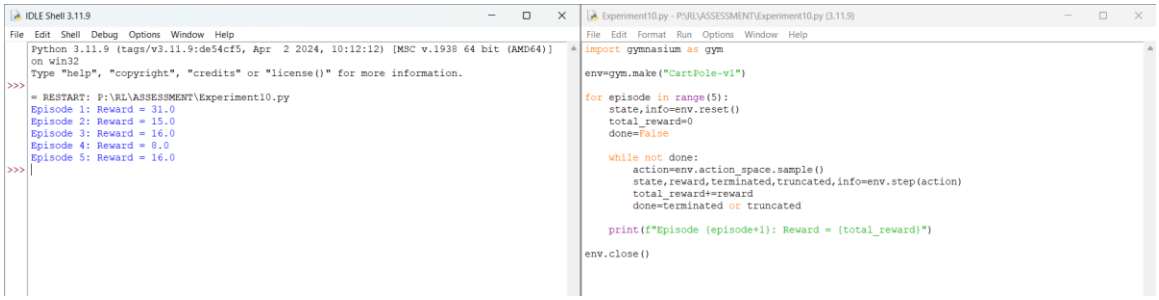
Title: Mini Reinforcement Learning Project Using CartPole

Aim: To develop a basic RL agent for the CartPole environment.

Algorithm:

• Import the required libraries.
• Initialize the environment/model.
• Execute the required RL procedure.
• Display the output.
• Verify the result.

Program & Output:



Result: The experiment was executed successfully and the expected output was obtained.